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The Dark Side of Generative Artificial Intelligence in Personal Learning: A Cognitive Perspective

Completed Research Paper

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Abstract

The increasing use of generative artificial intelligence (GenAI) in personal learning in higher education presents numerous challenges, highlighting the need for further research. This study explores the dark side of GenAI in personal learning from a cognitive perspective. Using thematic analysis of semi-structured interviews with university students, the study finds that creativity support, explainability, and self-regulated learning support directly shape learners' habitual GenAI use, with such relationships moderated by learners' trust in GenAI. The study reveals that habitual GenAI use can lead to critical thinking erosion, effort avoidance, cognitive automation, and cognitive deskilling, ultimately contributing to learners' cognitive atrophy. By highlighting the cognitive risks, this study advances the understanding of the dark side of GenAI use in personal learning and offers practical insights for promoting its sustainable use.

Keywords: Generative artificial intelligence; Dark affordances; Cognitive atrophy; Personal learning; Thematic analysis

Introduction

Generative artificial intelligence (GenAI) are intelligent systems capable of producing novel content and engaging in human-like dialogue (Krause *et al.*, 2025). Owing to its distinctive capabilities such as generating original outputs (Xia *et al.*, 2024), responding dynamically to prompts, and supporting creative generativity (Jin *et al.*, 2025), GenAI has become increasingly popular in higher education. As a result, understanding GenAI's role and its consequences for learners' cognitive processes has become important.

The growing use of GenAI is increasingly shaping personal learning in various manners. GenAI use, for example, influences learners' engagement by offering personalized support (Patil, 2025). It provides tailored explanations, assists with academic writing, and clarifies complex concepts (Naznin *et al.*, 2025), directly reinforcing habitual engagement in learning. By improving learning efficiency and reducing cognitive efforts, GenAI allows learners to focus on higher order thinking and deeper understanding (Shen *et al.*, 2025). Consequently, GenAI has become increasingly embedded in learning practices.

Much research has examined the consequences of GenAI use from diverse perspectives. Bai *et al.* (2023), for example, demonstrate that increased engagement with GenAI can lead to overreliance, thus diminishing learners' critical thinking capacity. Vera *et al.* (2025) find that the integration of artificial intelligence (AI) in STEM education affects learners' critical thinking and problem-solving abilities. Huang *et al.* (2025) report that higher intrinsic cognitive load negatively influences students' behavioral intention to use ChatGPT. Barcaui (2025) argues that unrestricted use of ChatGPT weakens learners' long-term memories. Collectively, these studies underscore the significant cognitive consequences of GenAI use in shaping personal learning from multiple perspectives.

Despite these insights, existing research remains fragmented. Limited attention has been paid to examine the dark side of GenAI use through learners' lived experiences, interaction patterns, and contexts of use. To address this research gap, this study answers the research question as follows: *What is the dark side of GenAI use in personal learning?*

This study explores the dark side of GenAI use in personal learning from a cognitive perspective. Using thematic analysis of semi-structured interviews with university students, the study finds that creativity support, explainability, and self-regulated learning support directly shape learners' habitual GenAI use, with such relationships moderated by learners' trust in GenAI. The study reveals that habitual GenAI use can lead to critical thinking erosion, effort avoidance, cognitive automation, and cognitive deskilling, ultimately contributing to learners' cognitive atrophy. By highlighting the cognitive risks associated with GenAI use, this study advances the understanding of the dark side of GenAI use in personal learning and offers practical insights for promoting its sustainable use.

The remainder of this paper is organized as follows. First, the related literature is reviewed, followed by a description of the research methodology. The findings are then presented and discussed. Finally, the paper concludes with the contributions, implications and limitations of the study.

GenAI Use in Personal Learning

GenAI is intelligent systems capable of generating human-like texts, images, code, and other content through large-scale generative models (Wood and Moss, 2024). It has increasingly being positioned as a transformative technology in personal learning due to its characteristics such as generativity and interactivity (Falebita and Kok, 2025), adaptability, and scalability (Essel *et al.*, 2024), which allow individual learners to access personal learning support across distinct learning tasks (Falebita and Kok, 2025). As a result, GenAI is becoming increasingly popular for enhancing personal learning.

Affordances are the possibilities that arise from the interaction between users and artifacts. They describe what can be achieved with a specific artifact in a given setting (Gibson, 1979). Technology affordances are about the potential use that digital technologies permit individuals to engage for specific goals (Majchrzak and Markus, 2013). They present the interactional nature of technology features, users' intentions, capabilities, and situational conditions (Majchrzak and Markus, 2013).

GenAI affordances are about the possibilities that arise from the collaboration between learners and GenAI (Majchrzak and Markus, 2013). They are related to various learning-related opportunities of GenAI in personal learning (Zhou *et al.*, 2025). These opportunities motivate learners to actively engage with learning content in flexible practices, thereby enhancing their personal learning.

GenAI plays a transformative role in shaping personal learning from different perspectives. GenAI, including large language models, AI chatbots, and adaptive tutoring systems, offers interactive and adaptive capabilities that reshape personal learning (Wood and Moss, 2024). It can generate human-like texts, reasoning, and problem-solving strategies (Krause *et al.*, 2025), shifting learning from static information retrieval to dynamic content creation and interaction. GenAI can support diverse cognitive tasks and adapt to user inputs (Essel *et al.*, 2024). It can accommodate diverse learning needs (Jin *et al.*, 2025), automate routine learning tasks, and provide real-time personalized feedback (Jin *et al.*, 2025), while enabling customized instructional content and context-aware guidance (McCarthy *et al.*, 2025). Jointly, these capabilities reduce learners' cognitive load for routine tasks and support problem-solving and metacognitive reflection, encouraging GenAI use in personal learning.

The increasing use of GenAI has raised concerns about the unintended consequence. While GenAI supports learning processes, excessive reliance may reduce learners' cognitive efforts (Bamasoud *et al.*, 2025), encourage superficial information processing (Essel *et al.*, 2024; Chen *et al.*, 2025), and weaken the development of learners' cognitive skills (Khan and Suhluli, 2025). GenAI use can steer learners toward outsourcing information processing, analytical reasoning, and problem-solving abilities (Chen *et al.*, 2025). Such dependence can amplify mental burden, contributing to cognitive atrophy (Khan and Suhluli, 2025).

There are several studies exploring the unintended consequence of GenAI use from different perspectives. Bai *et al.* (2023), for example, using a comprehensive survey, evaluate the impact of ChatGPT on human cognitive functions, revealing that increased engagement fosters over-reliance, resulting in reduced critical thinking capacity. Monib *et al.* (2024) examine the challenges of using Gen AI, showing that GenAI use is linked to reduced cognitive engagement. Barcaui (2025), adopting cognitive offloading theory, examines the impact of ChatGPT on knowledge retention, suggesting that unrestricted use impairs retention by reducing the cognitive effort necessary for durable memory. Patac and Patac (2025), using cognitive load theory, explore the effect of ChatGPT on students' cognitive load, showing that the detailed responses and efficient information retrieval lower intrinsic and extrinsic cognitive load, thus weakening learners' cognitive abilities. Bamasoud *et al.* (2025), integrating technology acceptance model and self-determination theory, investigate the cognitive factors shaping GenAI adoption, revealing that fear of overreliance, diminished student effort, and erosion of assessment credibility compromise individual adoption. Khan and Suhluli (2025), applying cognitive load theory, examine the relationship between cognitive demands and GenAI use in shaping research quality, showing that heightened cognitive load and fatigue associated with intensive GenAI use negatively affect the research outcome. Yang *et al.* (2025), using human-centered AI framework, explore the cognitive and emotional cost of GenAI use, highlighting that prompt uncertainty induces emotional fatigue, whereas response uncertainty triggers cognitive fatigue. Vera *et al.* (2025), adopting a design-based approach, investigate future STEM competencies, reporting that GenAI integration in STEM education affects learners' critical thinking and problem-solving abilities. Tian and Zhang (2025) examine university students' GenAI dependence on critical thinking, demonstrating that excessive reliance lowers learners' critical thinking, mediated by cognitive fatigue. These studies demonstrate that GenAI use generates adverse cognitive consequences that undermine personal learning.

The discussion above underscores that GenAI is shaping personal learning by influencing learners' cognitive processes through patterns of engagement. Although prior research has identified significant cognitive consequences of GenAI use (Bai *et al.*, 2023; Monib *et al.*, 2024), it is largely based on quantitative, review, and survey-based approaches that rely on predefined constructs and capture the outcomes rather than the underlying processes. While more advanced approaches have been proposed to operationalize and measure these outcomes (Du *et al.*, 2025), they remain focusing on measurement rather than explaining how such effects emerge through learners' lived experiences and engagement processes. Moreover, existing research often overlooks the contextual and behavioral dimensions of GenAI use, including how learners engage across different tasks, purposes, and levels of dependence. Consequently, how learners cognitively experience, interpret, and integrate GenAI into personal learning remains insufficiently understood. This highlights the need for a process-oriented exploration of learners' interactions with GenAI to better understand the mechanisms underlying these cognitive effects.

Research Methodology

This study adopts a qualitative research design using grounded theory to explore the cognitive consequences of GenAI use in personal learning. A qualitative approach is appropriate as it permits an in-depth understanding of learners' subjective experiences and process-oriented behaviors, which are difficult

to capture through predefined quantitative measures (Creswell and Creswell, 2018). Given the exploratory nature of this phenomenon, this approach allows for the examination of context-dependent learning processes and emergent cognitive consequences of GenAI use (Creswell and Creswell, 2018).

Grounded theory is appropriate for examining underexplored phenomena where existing theories inadequately explain emerging processes and behaviors (Charmaz, 2014). It enables theory development grounded in empirical data through iterative collection and analysis (Charmaz, 2014). In this study, grounded theory identifies patterns in learners' interactions with GenAI, focusing on how cognitive experiences and behavioral processes emerge across different contexts of use. Through constant comparison of codes and categories, the approach supports the development of a conceptual framework explaining how GenAI shapes personal learning outcomes. This makes it well suited for capturing the process-oriented and context-dependent nature of learner engagement with GenAI.

Semi-structured interviews are used to collect in-depth qualitative data on learners' experiences with GenAI in personal learning. This method supports flexible questioning, probing, and clarification, making it suitable for exploring complex and evolving phenomena (Chan and Hu, 2023). The interview protocol is informed by the study objectives and prior GenAI learning literature, covering patterns of GenAI use, cognitive experiences, behavioral engagement, and contextual influences. Open-ended questions and follow-up prompts are used to encourage detailed reflection and explore emerging issues in depth.

The interview consists of two sections. The first collects participants' demographic information, while the second explores their experiences, patterns of GenAI use, and perceptions of its influence on learning. To enhance content validity and clarity, the interview questions are reviewed by academic staff and higher degree research scholars prior to data collection. The questions are designed to be broad and exploratory, including prompts such as: *"What factors influence your use of GenAI?"*, *"What motivates you to use GenAI in learning?"*, *"What challenges do you face?"*, and *"How does GenAI influence your thinking or problem-solving?"* Interviews lasted approximately 30 minutes, are audio-recorded, and fully transcribed.

Recruitments are conducted using posters containing QR codes that invite interested students to submit an expression of interest to participate in the interview. The posters are distributed across relevant university channels to reach a broad range of students. The recruitment materials briefly outline the purpose of the study, the nature of participation, and the contact details for further information.

A three-step selection process is adopted to ensure rich, relevant, and diverse insights. First, students who have submitted the expression of interest are screened against the eligibility criteria, which require participants to be enrolled with prior experience using GenAI in personal learning. Second, eligible participants are contacted to schedule interviews. Third, an information and consent form is sent to confirmed students for their reading and signing, together with the interview link.

Purposive sampling is used to select learners who have prior experience using GenAI in personal learning. Snowball sampling has complemented this approach by enabling participants to refer peers with similar experiences, thereby broadening the range of viewpoints collected.

Twenty participants are recruited. Table 1 presents a summary of the demographic data. Of the twenty participants, sixteen are male and four are female. Most participants are doctoral students ($n = 14$), followed by master's students ($n = 3$) and undergraduate students ($n = 3$). Participants are represented in a range of academic disciplines, including Information Systems ($n = 5$), Engineering ($n = 4$), Management ($n = 3$), Supply Chain ($n = 3$), Applied Chemistry ($n = 2$), Health Science ($n = 1$), Architecture and Urban Planning ($n = 1$), and Economics ($n = 1$). Regarding the frequency of GenAI use, twelve participants have reported frequent use, seven have reported occasional use, and one has reported infrequent use. This variation reflects differing levels of familiarity and engagement with GenAI tools. Overall, the sample demonstrates diversity across educational levels, disciplines, and usage intensity, providing a broad and experience-rich perspective on learners' engagement with GenAI.

All interviews are conducted online via Microsoft Teams. Interviews are audio-recorded, transcribed verbatim, anonymized, and securely stored. Each interview begins with demographic questions, followed by open-ended questions exploring participants' experiences of GenAI use in personal learning. Question sequencing remains flexible to support natural conversational flow and encourage detailed reflection.

Table 1: Demographic characteristics of the participants

ID	Gender	Level of Education	Disciplines	Frequency Of Use
P1	Male	PhD	Management	Often
P2	Male	PhD	Information Systems	Always
P3	Male	PhD	Health Science	Often
P4	Male	PhD	Economics	Often
P5	Male	PhD	Architecture and Urban	Often
P6	Male	PhD	Applied Chemistry	Often
P7	Female	PhD	Information Systems	Often
P8	Male	PhD	Supply Chain	Always
P9	Male	PhD	Information Systems	Often
P10	Male	PhD	Engineering	Sometimes
P11	Female	PhD	Management	Often
P12	Male	PhD	Information Systems	Always
P13	Male	PhD	Applied Chemistry	Often
P14	Male	PhD	Management	Always
P15	Male	Bachelor	Engineering	Often
P16	Male	Bachelor	Engineering	Always
P17	Male	Bachelor	Engineering	Always
P18	Female	Master	Supply Chain	Often
P19	Female	Master	Information Systems	Always
P20	Male	Master	Supply Chain	Often

Thematic analysis is used for processing the interview data due to its flexibility, analytical rigor, and suitability for identifying the patterns across qualitative data (Braun and Clarke, 2021; Byrne, 2022). The use of this method supports a systematic examination of large volumes of textual data and facilitates the development of coherent themes that represent shared meanings across participants' accounts (Braun and Clarke, 2021). Its broad applicability makes thematic analysis particularly well suited for examining how GenAI use shapes personal learning from a cognitive perspective.

A six-phase thematic analysis procedure is followed for inductively deriving insights that are the representative of participants' responses, allowing the analysis to remain sensitive to participants' lived experiences of GenAI use (Braun and Clarke, 2021; Byrne, 2022). These phases include (a) familiarization with the data, (b) generation of initial codes, (c) collating codes into potential themes, (d) reviewing and refining themes, (e) defining and naming themes, and (f) organizing and reporting themes. First, the researchers familiarized themselves with the data through repeated reading of the transcripts, and meaningful segments of text are coded line-by-line to ensure fine-grained analysis. Second, initial codes are generated inductively to capture salient features of the data. Third, related codes are grouped into potential themes, which are iteratively refined through constant comparison across transcripts. Fourth, themes are reviewed to ensure internal coherence and external distinction. Fifth, themes are clearly defined and named. Finally, themes are organized and reported to provide a coherent account of participants' cognitive experiences, perceptions, and behaviors related to GenAI use.

Data analysis followed an iterative, inductive process (Braun and Clarke, 2021; Byrne, 2022). Open coding discovered important data segments, which are iteratively compared, refined, and sorted into higher-order categories through continual evaluation. These categories are then merged into themes indicating common patterns across participants, with constant alteration to guarantee consistency between codes and themes. NVivo 15 software is used to support data management, coding, and thematic development. To develop analytical rigor, the coding process is repeatedly reviewed by the research team, and relevant literature is accessed to contextualize the emerging themes.

Interview recordings are transcribed and reviewed to capture contextual nuances. Data collection and analysis continue until thematic saturation is reached, where no substantial new insights emerge. To enhance methodological rigor, coding is conducted iteratively and collaboratively, with codes compared and refined through discussion. This process improves consistency in theme development and strengthens analytical trustworthiness. A validation process is subsequently conducted to enhance the credibility of the emergent themes, consistent with qualitative research standards (Sabani *et al.*, 2023).

Triangulation is performed on the transcripts of participants P1 to P20. The validation and reliability of each theme require evidence from at least three participants through cross-checking within individual transcripts. For instance, the adverse effect of critical thinking erosion is confirmed through triangulation of the evidence from P1, P11, and P19. Table 2 summarizes the frequency of triangulated evidence.

Table 2: Summarizes the Frequency of the Triangulation	
Themes	Frequency of Triangulation Cumulative Sources
<i>GenAI-Supported Learning</i>	
Creativity Support	7
Explainability	4
Self-Regulated Learning Support	5
<i>Learner-Related Behavioral Mechanisms</i>	
Habitual GenAI Use	3
<i>Moderating Variables</i>	
Trust in GenAI	8
<i>Adverse Effects of GenAI Use</i>	
Erosion of Critical Thinking	5
Effort Avoidance	4
Cognitive Automation	3
Cognitive Deskilling	4

Research Findings

An inductive thematic analysis (Sabani *et al.*, 2019) is employed for investigating the dark side of GenAI in personal learning from the cognitive perspective. This method is selected because it allows for the discovery of subtle and unanticipated themes directly from participants' perception and experience.

The thematic analysis identified three themes: GenAI-supported learning, behavioral mechanisms, and adverse cognitive effects. In GenAI-supported learning, participants highlighted creativity support, explainability, and self-regulated learning support as key drivers of engagement with GenAI. Behavioral mechanisms captured habitual GenAI use as a process of repeated dependence, while trust in GenAI emerged as a moderating factor shaping learner engagement. Adverse cognitive effects included critical thinking erosion, effort avoidance, cognitive automation, and cognitive deskilling as negative consequences of GenAI use. These themes and subthemes are discussed below.

GenAI-Supported Learning

The integration of GenAI into personal learning is associated with the shift in how learners engage with learning. Participants have described GenAI's interactive, generative, and responsive characteristics as enabling them to use GenAI in their learning. As one participant P4 has noted, "*I find that the opportunities GenAI offers make me more confident in using it for my learning tasks, which encourages me to rely on it more regularly*". Within the data, three recurring sub-themes have been identified including creativity support, explainability, and self-regulated learning support.

Creativity support is related to the extent to which learners perceive GenAI as assisting idea generation and facilitating the completion of learning tasks, thereby providing effective support in personal learning (Zhou *et al.*, 2025). Learners regard GenAI as supportive because it offers alternative justifications, multiple perspectives, and linguistic enhancements (Zhou *et al.*, 2025). In personal learning where learners independently manage their learning (Peters and Angelov, 2025), GenAI is perceived as particularly valuable due to its immediate, responsive, and structured feedback. GenAI fosters creativity by helping learners overcome learning challenges through real-time idea generation, clarification, and structured

guidance, enabling more efficient and confident task engagement (Peters and Angelov, 2025). Learners rely on GenAI to scaffold brainstorm, organize thoughts, and shape academic expressions. They perceive GenAI as a creative partner in their learning. As Interviewee P1 has stated, *“It helps me to brainstorm ideas... it gives me so many viewpoints about whatever I am talking about.”* P12 has described GenAI as *“a helping hand to navigate writing difficulties like writer’s block,”* while P20 has noted that it *“helps to pull all the information together within minutes or even seconds.”* Participants have also emphasized linguistic refinement, with P4 has observed that *“GenAI helps shape grammar and tone for academic writing,”* and P6 has stated that it *“introduces new ways of expressing my ideas that I hadn’t previously considered.”* As a result, GenAI becomes embedded, reinforcing continued engagement in personal learning.

Explainability is related to the extent to which learners perceive GenAI as providing clear, contextualized, and tailored explanations that enhance comprehension and support knowledge acquisition in personal learning (Schneider, 2024). Learners value GenAI use because it breaks down complex concepts into manageable steps, presents multiple perspectives, and offers reasoning that is easy to follow (Schneider, 2024). In personal learning that requires learners to structure tasks, monitor understanding, and interpret information independently (Safarov et al., 2025), participants have reported that GenAI is particularly useful due to its immediate, structured, and responsive feedback, which helps them navigate conceptual challenges and consolidate understanding. Learners depend on GenAI to support reasoning, summarize content, and clarify difficult materials. They consider GenAI as an explanatory partner that complements and strengthens their independent cognitive efforts. As Interviewee P1 has demonstrated, *“When I face a challenge, I ask the GenAI... and it explains the whole thing like a mentor.”* P10 has reflected on its influence on engagement, stating, *“It explained it well for me... and now that idea has become a published paper.”* P8 has noted that GenAI *“summarizes the logic behind equations before I read the full article,”* and P14 has commented that *“GenAI breaks complex models into plain explanations that make sense.”* These accounts suggest that GenAI becomes integrated into learners’ routines, supporting repeated use for enhancing the development of comprehension, reasoning, and problem-solving skills.

Self-regulated learning support is about the degree to which learners perceive GenAI as enabling independent, personalized, and goal-oriented learning (Zhou et al., 2025). Learners describe GenAI as assisting them in planning, organizing, and managing learning activities, thereby providing structure aligned with their personal objectives (Lai, 2024). Features such as immediate feedback, continuous responsiveness, and easy access (Belhaj, 2025) are emphasized for allowing learners to monitor progresses and take responsibilities for their developments. In personal learning that promotes self-directed engagement (Lai, 2024), learners use GenAI as a supportive mechanism for sustaining independent learning. They rely on GenAI to support self-monitoring and confirmatory evaluation. As Interviewee P7 has stated, *“I use it to check my understanding before submitting my assignment,”* while P9 has explained, *“When I don’t understand something, I try first, then I use GenAI to confirm whether I am correct.”* P1 has noted, *“I can evaluate my own answer by comparing it with what GenAI provides.”* Others have also highlighted time management and self-testing, with P5 has observed, *“It helps me manage my time because I can get quick clarification when I’m stuck,”* and P14 has stated, *“I use it to test myself by asking questions related to the topic.”* These descriptions signal that GenAI functions as a regulatory support in personal learning, enabling learners to set goals, monitor progress, and adjust strategies in their learning processes. Table 3 summarizes these findings.

Behavioral Mechanisms

Behavioral mechanisms reflect learners’ engagement with GenAI, translating positive perceptions into sustained use. Participants note that frequent interaction reinforces confidence in GenAI outputs. This repeated engagement fosters habitual use, which may introduce adverse cognitive effects. As one interviewee P3 has noted, *“I am using it quite often... I often use GenAI”*. Within the interview data, one persistent sub-theme has been identified, including habitual GenAI use.

Habitual GenAI use is about the extent to which learners perceive GenAI as enabling consistent and automatic engagement with cognitive, explanatory, and decision-making tasks, thereby supporting personal learning (Sousa and Cardoso, 2025). Unlike occasional or situational use, habitual use is important because it reflects the routinization of GenAI within learners’ everyday learning practices, shaping how they approach reasoning and problem-solving tasks (Moradi, 2025). Learners described GenAI as facilitating repeated interactions through instant feedback, personalized guidance, and continuous accessibility (Moradi, 2025), which reinforce reliance on the tool. Participants have frequently

characterized this pattern as “automatic” (P3), “default” (P7), and “assumed” in task planning (P10). For instance, P3 has noted, “*I just open it automatically when I start.*” Participants have distinguished habitual use from other forms of engagement through differences in intentionality and cognitive engagement prior to use (Sousa and Cardoso, 2025). Learners reporting early or limited use have described more deliberate decision-making, such as P1 who has stated, “*At first I was deciding if it was appropriate.*” In contrast, habitual users have described a shift in focus from deciding whether to use GenAI to evaluating its outputs, as P6 has explained, “*Now I focus on fixing the output, not deciding to use it.*” These different usage patterns are associated with different learning effects. Habitual users link GenAI to conceptual support, with P12 has noted, “*It helps me understand things before I try myself,*” whereas situational users frame their use primarily in terms of efficiency, such as P1 has stated, “*It just saves time,*” or P19 has explained, “*I use it mainly when I’m stuck.*” In personal learning where learners independently manage learning tasks and interpret information, GenAI is perceived as supporting reasoning by allowing learners to offload cognitive effort while maintaining engagement with learning activities (Wu et al., 2025). Participants have also expressed concerns that habitual use may gradually reduce independent reasoning. As P1 has observed, “*Instead of your thinking, you’re allowing GenAI to think for you,*” while P10 has noted that learners “*immediately put it to ChatGPT without even, for a second, thinking about it.*” These findings highlight the significance of habitual GenAI use, its distinction from other use patterns, and the different ways in which routine interactions with GenAI shape learners’ reasoning, decision-making, and independent cognitive engagement.

Table 3: Themes and Findings on GenAI-Supported Learning

Themes	Description	Supported Quotes
Creativity support	Related to the extent to which learners perceive GenAI as assisting idea generation and facilitating the completion of learning tasks	P1: “ <i>It helps me to brainstorm ideas... it gives me so many viewpoints about whatever I am talking about.</i> ” P12: “ <i>a helping hand to navigate writing difficulties like writer’s block,</i> ” P20: “ <i>helps to pull all the information together within minutes or even seconds.</i> ” P4: “ <i>GenAI helps shape grammar and tone for academic writing,</i> ” P6: “ <i>introduces new ways of expressing my ideas that I hadn’t previously considered.</i> ”
Explainability	Related to the extent to which learners perceive GenAI as providing clear, contextualized, and tailored explanations that enhance comprehension and support knowledge acquisition in personal learning	P1: “ <i>When I face a challenge, I ask the GenAI... and it explains the whole thing like a mentor.</i> ” P10: “ <i>It explained it well for me... and now that idea has become a published paper.</i> ” P8: “ <i>summarizes the logic behind equations before I read the full article,</i> ” P14: “ <i>GenAI breaks complex models into plain explanations that make sense.</i> ”
Self-regulated learning support	The degree to which learners perceive GenAI as enabling independent, personalized, and goal-oriented learning	P7: “ <i>I use it to check my understanding before submitting my assignment,</i> ” P9: “ <i>When I don’t understand something, I try first, then I use GenAI to confirm whether I am correct.</i> ” P1: “ <i>I can evaluate my own answer by comparing it with what GenAI provides.</i> ” P5: “ <i>It helps me manage my time because I can get quick clarification when I’m stuck,</i> ” P14: “ <i>I use it to test myself by asking questions related to the topic.</i> ”

The moderating variable is related to the factor that shapes how learners interact with GenAI. Participants have described this primarily through trust in GenAI, which influences the strength and effectiveness of their engagement across different learning situations. As one interviewee P19 has noted, “*I feel confident in using GenAI without any support*”, illustrating how trust can strengthen learners’ reliance on and interaction with GenAI in their learning processes.

Trust in GenAI is related to the extent to which learners perceive GenAI as reliable and worth engaging with in facilitating the learning process (Wang *et al.*, 2025). Learners' trust in GenAI shapes how they interact with, rely on, verify, and regulate its use, as well as their confidence in integrating it into personal learning (Lelescu *et al.*, 2025), highlighting its role as a moderating component in learners' engagement. Several participants have portrayed trust in GenAI as evolving over time. For example, Interviewee P7 has explained, *"I trust it more now because it's been right before,"* while P11 has noted, *"The trust built up over time,"* linking this progression to more regular use. Other participants have deliberately confined their trust to specific learning tasks, as P4 has stated, *"I trust it for explanations, not final answers,"* and P16 has asserted, *"I wouldn't trust it for assessment work."* These accounts indicate that trust moderates how learners translate GenAI's support into learning actions, shaping the extent to which they rely on or verify its outputs. In personal learning contexts where learners independently navigate content, evaluate information, and make decisions, trust enables them to engage with GenAI while maintaining oversight of its responses (Lelescu *et al.*, 2025). Learners recognize GenAI as dependable in certain contexts, which reinforces repeated interaction while also encouraging cautious engagement. As Interviewee P2 has confirmed, *"I double-check what GenAI tells me because I've seen mistakes, but mostly it's reliable."* P7 has noted, *"I've learned to trust, but verify when it's consistent with my reading, then I rely on it."* P11 has reflected, *"With experience, I know when to believe it and when to question it,"* while P9 has emphasized, *"It sometimes gives references that don't exist, and that makes me cautious."* Collectively, participants' descriptions suggest that trust in GenAI develops through experience and moderates how GenAI support is incorporated into routine learning rather than directly determining habitual use. Table 4 summarizes these findings.

Adverse Cognitive Effects

Increased interactions with GenAI are accompanied by growing perceptions of cognitive offloading, which over time may lead to unintended cognitive effects. Participants have expressed how their routine and sustained use of GenAI leads to emerging consequences in learning. These effects illustrate how repeated engagement can reduce learners' critical thinking, problem-solving, creativity, and independent reasoning, potentially resulting in cognitive atrophy (Hassen, 2025).

Cognitive atrophy is about how learners feel their thinking skills including critical thinking, problem-solving, creativity, and independent reasoning gradually diminish due to excessive reliance on GenAI (Hassen, 2025). As one participant P19 has observed, *"thinking power... is being diminished by GenAI,"* and *"critical thinking... has been reduced"*. Within the data, four sub-themes have been identified, including critical thinking erosion, effort avoidance, cognitive automation, and cognitive deskilling.

Critical thinking erosion is related to the extent to which learners perceive GenAI use as reducing independent reasoning, evaluative judgment, and analytical engagement (Tian and Zhang, 2025). It is evident when learners accept GenAI outputs with limited critical evaluation in tasks requiring justification, driven by repeated reliance on ready-made explanations, solutions, and decisions. In personal learning, where autonomous reasoning and sustained cognitive effort are essential (Vera *et al.*, 2025), such reliance may weaken evaluative capacity and independent thinking, contributing to cognitive atrophy. Although GenAI can enhance efficiency and provide immediate guidance (Chen *et al.*, 2025), frequent unregulated use may shift cognitive responsibility away from learners, reinforcing passive engagement and constraining deep critical thinking. As Interviewee P10 has observed, *"Learners immediately put it to ChatGPT without even just for a second thinking about it"*. P10 further has noted that *"critical thinking and decision-making... has been reduced as well."* P1 has stated, *"Instead of your thinking, you're allowing the GenAI to think for you,"* highlighting a diminished tendency to independently evaluate information. These accounts denote that ready-access to GenAI-generated outputs may prevent deep analytical engagement and active problem-solving, progressively weaken learners' independent cognitive capacities and contribute to the rise of cognitive atrophy.

Effort avoidance is about the degree to which learners perceived GenAI as reducing the cognitive struggle required to engage deeply with learning (Spitzer *et al.*, 2025). Learners describe GenAI as providing immediate, structured, and simplified solutions, which lower the mental effort needed for problem-solving and reasoning. In personal learning where persistence, reflection, and sustained effort are critical for development (Andewi *et al.*, 2025), such convenience can reduce learners' willingness to tackle challenging tasks independently. Given that GenAI reduces cognitive struggle and supports task completion (Huang *et al.*, 2025), learners verify that reliance on its ready-made outputs fosters dependence on quick solutions,

bypassing opportunities for active reasoning and deeper engagement. Accordingly, learners perceive GenAI as a tool that eases cognitive load. As explained by Interviewee P1, *“Instead of doing things the hard way, you want it the easy way,”* while P10 has questioned, *“Why do we have to struggle... when GenAI is there?”* P17 has further emphasized, *“It gives everything to you on your plate.”* These explanations suggest that reliance on quick solutions can restrain sustained cognitive efforts, reinforce effort-avoidant behaviors, and promote passive engagement with challenging tasks, thereby contributing to cognitive atrophy.

Table 4: Themes and Findings on Habitual GenAI Use and Trust in GenAI		
Themes	Description	Supported Quotes
Habitual GenAI use	The extent to which learners perceive GenAI as enabling consistent and automatic engagement with cognitive, explanatory, and decision-making tasks	P3: <i>“I just open it automatically when I start”</i> P10: <i>“I plan my assignments knowing I’ll use it”</i> P1: <i>“At first I was deciding if it was appropriate”</i> P6: <i>“Now I focus on fixing the output, not deciding to use it”</i> P12: <i>“It helps me understand things before I try myself”</i> P1: <i>“It just saves time”</i> or P19: <i>“I use it mainly when I’m stuck”</i> P1: <i>“Instead of you thinking, you’re allowing the GenAI to think for you.”</i> P10: <i>“Learners immediately put it to ChatGPT without even, for a second, thinking about it,”</i> P10: <i>“critical thinking and decision-making... has been reduced as well.”</i>
Trust in GenAI	The extent to which learners perceive GenAI as reliable and worth engaging with in facilitating learning process	P7: <i>“I trust it more now because it’s been right before”</i> P11: <i>“The trust built up over time”</i> P4: <i>“I trust it for explanations, not final answers”</i> P16: <i>“I wouldn’t trust it for assessment work”</i> P2: <i>“I double-check what GenAI tells me because I’ve seen mistakes, but mostly it’s reliable.”</i> P7: <i>“I’ve learned to trust, but verify when it’s consistent with my reading, then I rely on it.”</i> P11: <i>“With experience, I know when to believe it and when to question it,”</i> P9: <i>“It sometimes gives references that don’t exist, and that makes me cautious.”</i>

Cognitive automation is about the extent to which learners perceive GenAI as a default cognitive partner that performs thinking-related tasks by generating immediate, structured, and seemingly reliable responses (Engel *et al.*, 2022). It is particularly evident in routine or time-constrained tasks where learners rely on GenAI for reasoning, problem-solving, and decision-making. In personal learning, which requires autonomous sense-making and reflective thinking (Patil, 2025; Spitzer *et al.*, 2025), such automation may reduce opportunities for active knowledge construction. Over time, reliance on GenAI can substitute learners’ own reasoning and effort, diminishing active cognitive engagement. As a result, learners perceive GenAI as a cognitive shortcut that reduces the need for sustained thinking. As Interviewee P17 has described, learners enter an *“autopilot mode... GenAI does everything for you.”* P19 has noted reliance on *“one system instead of relying on multiple sources,”* while P9 has described prompting GenAI *“without even just for a second thinking,”* reflecting task-level substitution of cognitive effort. These justifications suggest that frequent delegation of thinking and problem-solving to GenAI strengthens cognitive automation, thereby deteriorating independent cognitive performance and initiating a pathway toward cognitive atrophy.

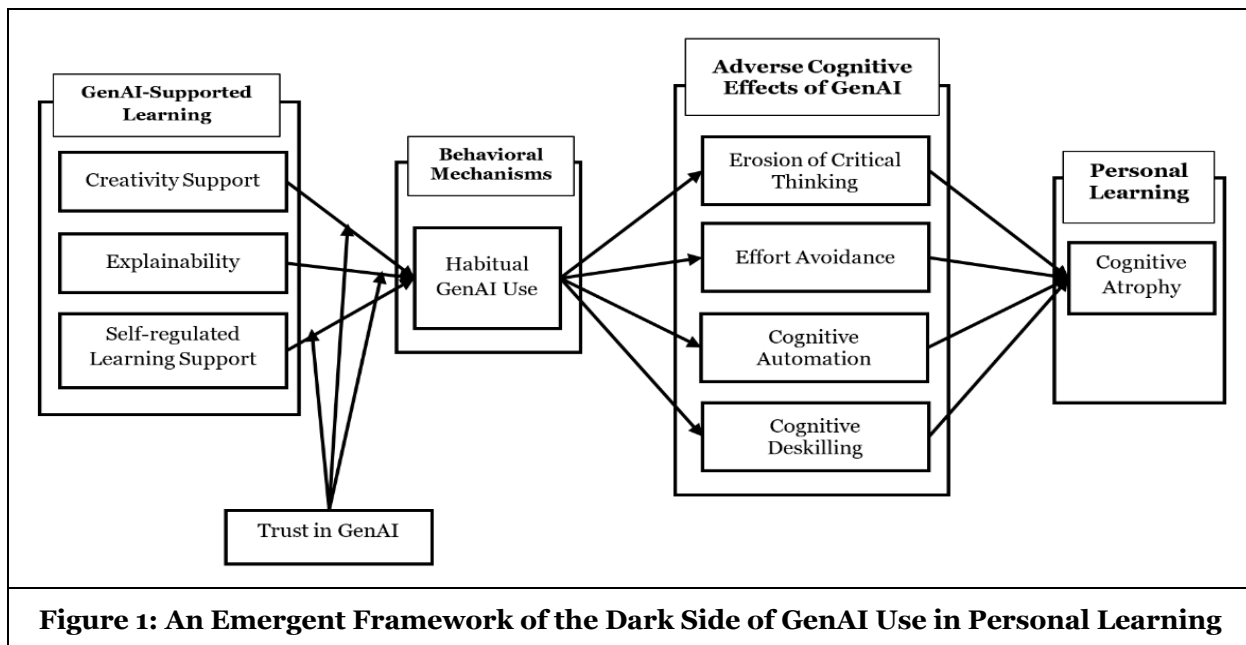
Cognitive deskilling is related to the extent to which learners perceive sustained use of GenAI as gradually reducing their need to actively exercise and strengthen their own cognitive capacities, such as idea generation, content analysis, and decision-making. (Lu and Hu, 2025). This is remarkably evident in sustained dependence on GenAI across various learning-related tasks, where independent cognitive practices become increasingly diminished. Learners report that frequent reliance on GenAI shifts cognitive

skills from themselves to the system, limiting opportunities for deep reasoning, critical reflection, and active knowledge construction (Vera *et al.*, 2025). In personal learning where learning autonomy, self-skill development, and active engagement are central, such a shift fosters superficial interactions with content, constraining self-directed intellectual developments and deliberate skill-building (Bai *et al.*, 2023; Li, 2023). Although GenAI enhances analytical reasoning and skill development, learners report that sustained dependence can diminish opportunities to develop learners' own cognitive skills, shifting ownership, authenticity of GenAI-assisted outputs, and sense-making from learners to GenAI. As Interviewee P1 has questioned, *"Is that introduction my work or the work of the GenAI?"* demonstrating blurred authorship and diminished cognitive ownership. P20 further has observed, *"You don't really know what you're putting together,"* highlighting the decline in learners' ability to independently construct and understand their work. These interpretations clarify that transferring responsibility for cognitive tasks to GenAI emphasizes a path toward weakened independent cognitive functioning, thereby accelerating cognitive atrophy. Table 5 summarizes these findings.

Table 5: Themes and Findings on Adverse Cognitive Effects		
Themes	Description	Supported Quotes
Critical Thinking Erosion	Related to the extent to which learners perceive GenAI as diminishing the need for independent reasoning, evaluative judgment, and analytical engagement	P10: <i>"Learners immediately put it to ChatGPT without even just for a second thinking about it,"</i> P10: <i>"critical thinking and decision-making... has been reduced as well."</i> P1: <i>"Instead of you thinking, you're allowing the GenAI to think for you,"</i>
Effort Avoidance	The degree to which learners perceive GenAI as reducing the cognitive struggle required to engage deeply with learning	P1: <i>"Instead of doing things the hard way, you want it the easy way,"</i> P10: <i>"Why do we have to struggle... when GenAI is there?"</i> P17: <i>"It gives everything to you on your plate."</i>
Cognitive Automation	The extent to which learners perceive GenAI as a default cognitive partner that performs thinking tasks on their behalf, providing immediate and seemingly reliable responses	P17: <i>"autopilot mode... GenAI does everything for you."</i> P19: <i>"one system instead of relying on multiple sources,"</i> P9: <i>"without even just for a second thinking,"</i>
Cognitive Deskilling	Related to the extent to which learners perceive GenAI as reducing the need to actively exercise and strengthen their own cognitive capacities	P1: <i>"Is that introduction my work or the work of the GenAI?"</i> P20: <i>"You don't really know what you're putting together,"</i>

The discussion above highlights the dark side of GenAI in personal learning. Each adverse cognitive effect signifies a specific cognitive consequence. Collectively they reveal a broader and more concerning outcome. Participants' experiences show that sustained dependence on GenAI can decrease evaluative depth, reduce persistence in difficult problem-solving, transfer core reasoning functions to automated outputs, and progressively weaken independently practiced skills. If left unaddressed, such consequences can generate significant cognitive effects, thereby contributing to cognitive atrophy (Hassen, 2025). This process develops collectively through habitual and trust-driven reliance on GenAI, where accessibility and effectiveness steadily diminish sustained intellectual efforts. This blend of unintended consequences provides a foundation for understanding how these adverse cognitive effects are derived from learners habitual GenAI use in personal learning.

The thematic analysis indicates that GenAI shapes personal learning through a dynamic interplay between enabling conditions, behavioral mechanisms, and emerging cognitive consequences. Participants have highlighted that creativity support, explainability, and self-regulated learning support create enabling conditions that facilitate learners' engagement with GenAI. These conditions influence how learners interact with GenAI across different contexts of use, which in turn shape behavioral mechanisms such as habitual use and trust formation. These habitual patterns reinforce cognitive reliance on GenAI, reshaping how learners approach learning tasks and decision-making. The findings further show that these behaviors are not uniform but evolve through repeated, context-dependent interactions. Sustained and intensive reliance may give rise to adverse cognitive consequences, including critical thinking erosion, effort avoidance, cognitive automation, and cognitive deskilling. Importantly, these effects emerge through learners' engagement processes rather than from GenAI use in isolation. These insights are synthesized into a conceptual framework, shown as in Figure 1, which describes GenAI use in personal learning as a progression from enabling conditions, through behavioral mechanisms, to cognitive consequences. The framework illustrates how these components interact over time to produce varied learning outcomes, ultimately contributing to cognitive atrophy under sustained overreliance.



Discussion

Prior studies on cognitive atrophy have begun to operationalize the concept using qualitative approaches to capture its behavioral and experiential manifestations, which are often overlooked in measurement-based designs (Du *et al.*, 2025). Building on this line of inquiry, this qualitative study explores the dark side of GenAI use in shaping learners' cognitive experiences in personal learning. The findings show that GenAI-supported learning where creativity support, explainability, and self-regulated learning guidance shape how learners approach tasks contribute to habitual GenAI use and trust in GenAI. Furthermore, the study reveals that habitual use and trust can generate adverse cognitive effects, including critical thinking erosion, effort avoidance, cognitive automation, and cognitive deskilling, thereby contributing to cognitive atrophy.

In the theme "*GenAI-supported learning*," participants have emphasized how creativity support, explainability, and self-regulated learning guide their daily learning routines. Learners describe using GenAI to generate ideas, clarify complex concepts, and organize tasks independently. These experiences suggest that GenAI-supported learning is closely interconnected with learners' personal goals and habits. Evidence indicates that learners' perceptions of these supports motivate repeated and routine engagement with the tool for learning tasks. This aligns with prior research showing that creativity support and explainability can stimulate idea generation, enhance comprehension, and promote self-directed learning (Lai, 2024; Peters and Angelov, 2025), leading to habitual GenAI use in supporting their learning.

In the theme “*behavioral mechanisms*,” repeated engagement and growing confidence with GenAI is emerged as central to shaping learning behaviors. Habitual GenAI use is emerged as a salient aspect, with participants describing it as “automatic,” “default,” and “assumed” in learning-related tasks and planning. Participants have distinguished habitual use from other usage patterns and highlighted variations in learners’ experiences through which GenAI becomes routinely embedded in everyday learning practices. Many participants have reflected that these supports encourage repeated interaction, reinforcing perceptions of reliability and comfort in integrating GenAI into learning activities. These observations are consistent with research suggesting that sustained interaction with GenAI tools can strengthen learners’ confidence and promote habitual use (Salifu *et al.*, 2025).

In the “*moderating variable*,” individual learners’ contextual and behavioral factors shape how they engage with and respond to GenAI, thereby strengthening and directing the relationship between GenAI-supported learning and behavioral mechanisms across different learning situations. Trust in GenAI is emerged as a moderating factor, shaping how GenAI support is taken up and incorporated into everyday learning processes instead of directly driving habitual GenAI use. Participants have described trusting in GenAI for cognitive and decision-making support, often delegating tasks that they would otherwise complete independently. Trust has also amplified habitual use, aligning with studies highlighting trust as a key mediator in human-AI collaboration, where reliability perceptions drive consistent reliance (Wang *et al.*, 2025). However, participants’ accounts indicate that habitual engagement may gradually shift cognitive responsibility away from learners themselves, reflecting the literature on cognitive offloading and the potential for dependency when GenAI is repeatedly relied upon (Sousa and Cardoso, 2025; Wu *et al.*, 2025) which in turn results in unintended adverse cognitive effects.

In the theme “*adverse cognitive effects*,” participants have reported that sustained reliance on GenAI contributes to the erosion of critical thinking, effort avoidance, cognitive automation, and cognitive deskilling. They have described diminished engagement with challenging tasks, reduced creativity, and weaker independent reasoning. These reflections align with prior research indicating that overreliance on GenAI can reduce cognitive effort, encourage passive learning, and undermine skill development (Tian and Zhang, 2025; Vera *et al.*, 2025). Participants have also noted that effortless access to GenAI responses often promotes cognitive offloading. These adverse cognitive consequences operate at several levels of learning and reflect a range of learners’ cognitive processes. Critical thinking erosion, for example, reflects an immediate reduction in evaluative engagement during task completion. Effort avoidance captures learners’ tendency to minimize the cognitive struggle required for deep learning. Cognitive automation reflects the task-level substitution of thinking processes by GenAI, whereas cognitive deskilling signifies a longer-term reduction in learners’ independent cognitive capability resulting from sustained reliance.

These findings show that the adverse cognitive effects of GenAI are not uniform but are shaped by the context of use and learners’ interaction patterns. Specifically, task type, frequency of use, and dependence behaviors influence how GenAI affects cognitive processes. For example, repetitive or time-constrained tasks tend to encourage cognitive automation, whereas evaluative or reflective tasks are more likely to expose critical thinking erosion when GenAI outputs are accepted without scrutiny. Over time, sustained reliance on GenAI across learning contexts may contribute to cognitive atrophy.

Grounded in these findings, a conceptual framework is developed to explain the dark side of GenAI use in personal learning. The framework shows that cognitive atrophy is not a direct outcome of GenAI affordances but emerges through a mediated process in which GenAI-supported learning fosters habitual use shaped by trust built through repeated interactions. While these mechanisms initially enhance cognitive performance, sustained reliance on GenAI can lead to adverse effects, including critical thinking erosion, effort avoidance, cognitive automation, and cognitive deskilling. Overall, the framework highlights the interplay between GenAI-supported learning, behavioral mechanisms, and cognitive consequences, showing how repeated, trust-driven engagement can result in long-term cognitive atrophy.

Contributions and Implications

Theoretical Contributions

This study offers three major theoretical contributions. First, it provides a structured framework for understanding how GenAI-supported learning translates into adverse cognitive effects through habitual use. By operationalizing key constructs including GenAI-supported learning, habitual GenAI use, trust in

GenAI, adverse cognitive effects, and cognitive atrophy. It explains how enabling technologies can progressively weaken learners' cognitive engagement. Unlike prior studies that focus on positive outcomes such as performance and self-regulated learning, this study conceptualizes the cognitive consequences of prolonged, uncritical use, identifying cognitive atrophy as a key yet underexplored effect of habitual GenAI dependence.

Second, the study integrates the notion of the dark side of GenAI at the learners' cognitive level. Rather than assessing GenAI exclusively as a cognitive augmentation tool, it examines how habitual reliance can shift GenAI's role from a supportive framework to an operative replacement of core cognitive processes. This shift provides a theoretically grounded rationale for understanding how critical thinking ability, problem-solving capacity, and creativity may diminish over time. By articulating these restraining pathways, the study facilitates a more holistic understanding of the inhibitory dynamics of GenAI-enabled learning.

Finally, the study offers a generalizable framework, making the proposed model suitable for comparative and empirical validation across diverse contexts. Its descriptive transparency permits future researchers to systematically investigate the strength, direction, and boundary conditions of the relationships between the proposed constructs in personal learning.

Practical Implications

The study offers numerous immediate and actionable implications. First, it highlights the importance of considering the dark sides of GenAI in fostering responsible and reflective integration. By identifying cognitive-constraining effects, the study assists educators and university leaders in understanding GenAI's role in learning and supports more informed decisions regarding instructional design, assessment structures, and technology governance for sustainable and cognitively engaging learning.

Second, the study emphasizes the importance of balancing GenAI use with active cognitive engagement. Educators can design pedagogical strategies that position GenAI as a complementary cognitive scaffold rather than a substitute for learner effort. This includes, for instance, developing structured prompting exercises, reflective reasoning on GenAI-generated responses, and staged problem-solving activities that involve independent reasoning before GenAI consultation. Such methods can alleviate over-reliance and decrease the risk of cognitive atrophy.

Third, universities can leverage the findings and framework to develop institutional guidelines that encourage responsible GenAI use. These guidelines should highlight GenAI literacy, metacognitive awareness, and critical evaluation skills, ensuring that learners remain cognitively engaged while benefiting from GenAI-supported learning.

Finally, policymakers and curriculum designers can use the findings to guide the strategic integration of GenAI within educational settings. By supporting GenAI adoption with objectives such as deep learning, creativity, and independent thinking, institutions can maximize pedagogical benefits while minimizing unintended cognitive consequences. The findings and proposed framework are directly applicable for guiding policy development, instructional innovation, and institutional strategies aimed at sustaining meaningful personal learning in GenAI-supported environments.

Conclusion

This study examines the cognitive risks of GenAI in personal learning. GenAI-supported features such as creativity enhancement, explainability, and self-regulated learning support encourage habitual engagement, while trust conditions continued reliance. Over time, habitual GenAI use may foster dependence, leading to erosion of critical thinking, effort avoidance, cognitive automation, and deskilling. These unintended effects can weaken deep processing, independent reasoning, and sustained cognitive engagement, potentially resulting in cognitive atrophy. The study develops a conceptual framework for explaining how GenAI may constrain cognition and provides guidance for educators, universities, and policymakers to promote responsible GenAI use while preserving learner autonomy and reflective capacity.

Several limitations warrant consideration for future research. The qualitative design and the relatively high proportion of PhD students limit generalizability, although their extensive engagement with advanced tasks makes them suitable for examining cognitive-level processes. Future research should include more diverse and cross-disciplinary samples. The study also does not quantify effect magnitudes or capture long-term

consequences, highlighting the need for longitudinal, experimental, and mixed-method approaches to examine cognitive atrophy (Du *et al.*, 2025). As GenAI effects are context-dependent, future work should explore how different usage patterns, instructional interventions, and social or collaborative settings shape its cognitive consequences. The framework is developed from qualitative analyses of participants' accounts and is therefore exploratory in nature, indicating an interpretative synthesis instead of an entirely validated theoretical model. Future research should empirically test and enhance it using quantitative or mixed-method approaches.

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